Embedded Systems Final Project

Smart Real-Time Driver Monitoring & Emergency Detection

Addis Ababa Science and Technology University

Tesfamichael Tafere

Tewodros Gebretsadik

Thressa Menilek

Tigist Kinfemichael

Tsegaye Berhe

Submitted to: Kassahun Tadesse

MISSION & MOTIVATION

The Challenge

Road traffic accidents claim **1.19 million lives** annually. Fatigue and micro-sleep are major preventable contributors.

The Mission

Develop a unified system that monitors driver state **preventively** and manages crash response **reactively**.

Core Integration

- Computer Vision for state monitoring
- Embedded sensors for crash detection
- GPS location for emergency tracking
- Multi-tier alert system (Buzzer/LED)

CURRENT SAFETY GAPS

Monitoring Gap

Lack of early-state detection in older or low-cost vehicles for drowsiness and distraction.

Response Lag

Delayed emergency response when drivers are incapacitated and unable to call for help.

System Fragmentation

Monitoring and crash reporting are usually separate, increasing complexity and cost.

Cost Barriers

Commercial systems are often expensive and difficult to retrofit into existing fleets.

TWO-TIER SAFETY STRATEGY

Tier 1: Preventive Monitoring

Continuous real-time eye and head posture analysis using computer vision to detect early fatigue.

- ✓ Drowsiness Detection
- ✓ Yawning & Alertness
- ✓ Head Pose Analysis

Tier 2: Reactive Detection

Sensor-based detection for critical impacts or sustained critical driver failure.

- ✓ Crash-like Acceleration
- ✓ GPS Coordinate Capture
- ✓ Automatic SOS Generation

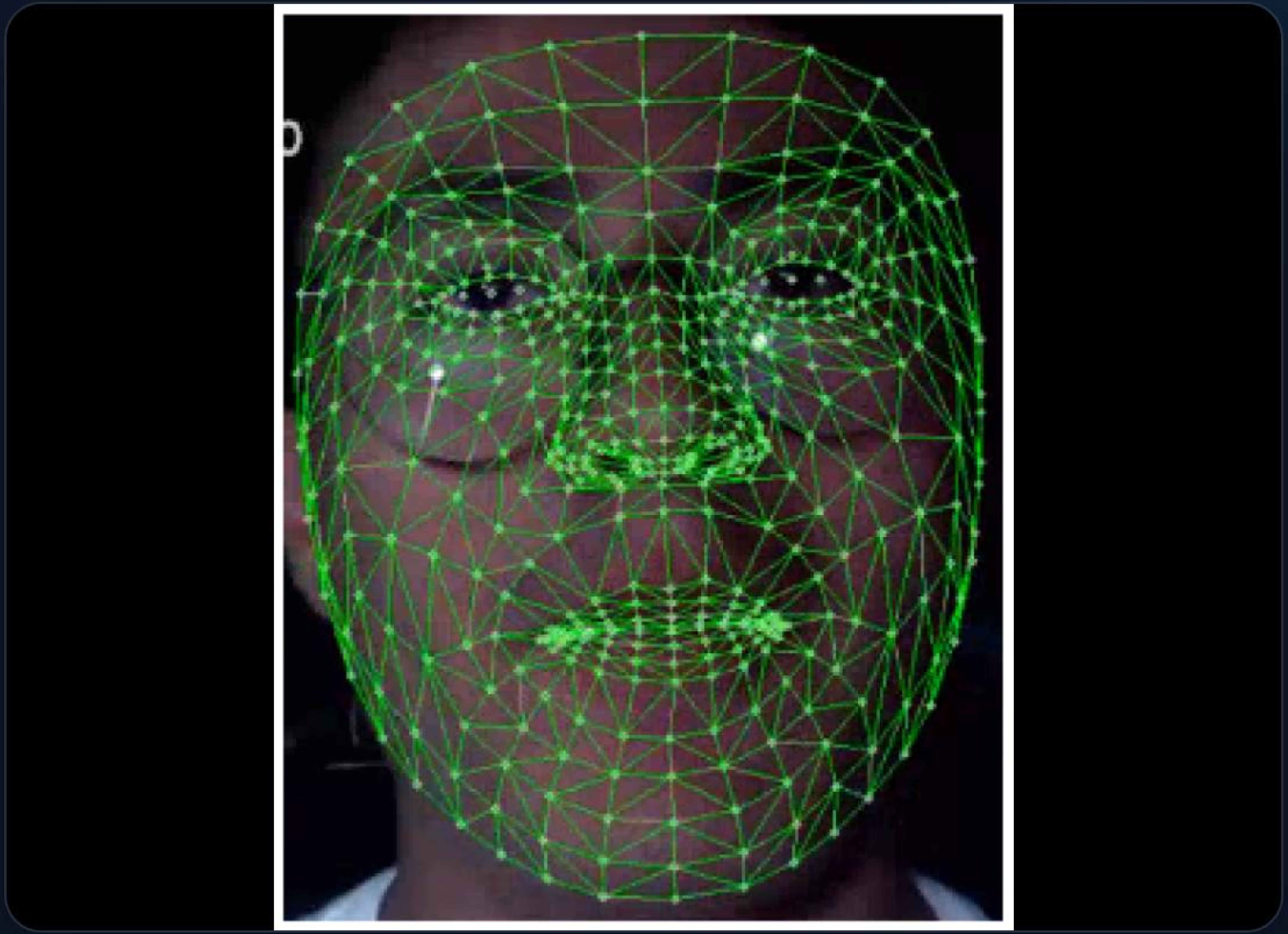
SYSTEM ARCHITECTURE

1. Perception: Camera captures video, OpenCV extracts facial landmarks.

2. Analysis: Fatigue score (0-100) calculated based on EAR/PERCLOS.

3. Processing: ESP32 State Machine evaluates sensor inputs + score.

4. Output: Local LCD/Buzzer alerts or GSM/GPS SOS message.



COMPUTER VISION METRICS

EAR & PERCLOS

Measures eye aspect ratio and closure percentage to identify drowsiness.

MAR

Mouth aspect ratio tracks yawning frequency for fatigue estimation.

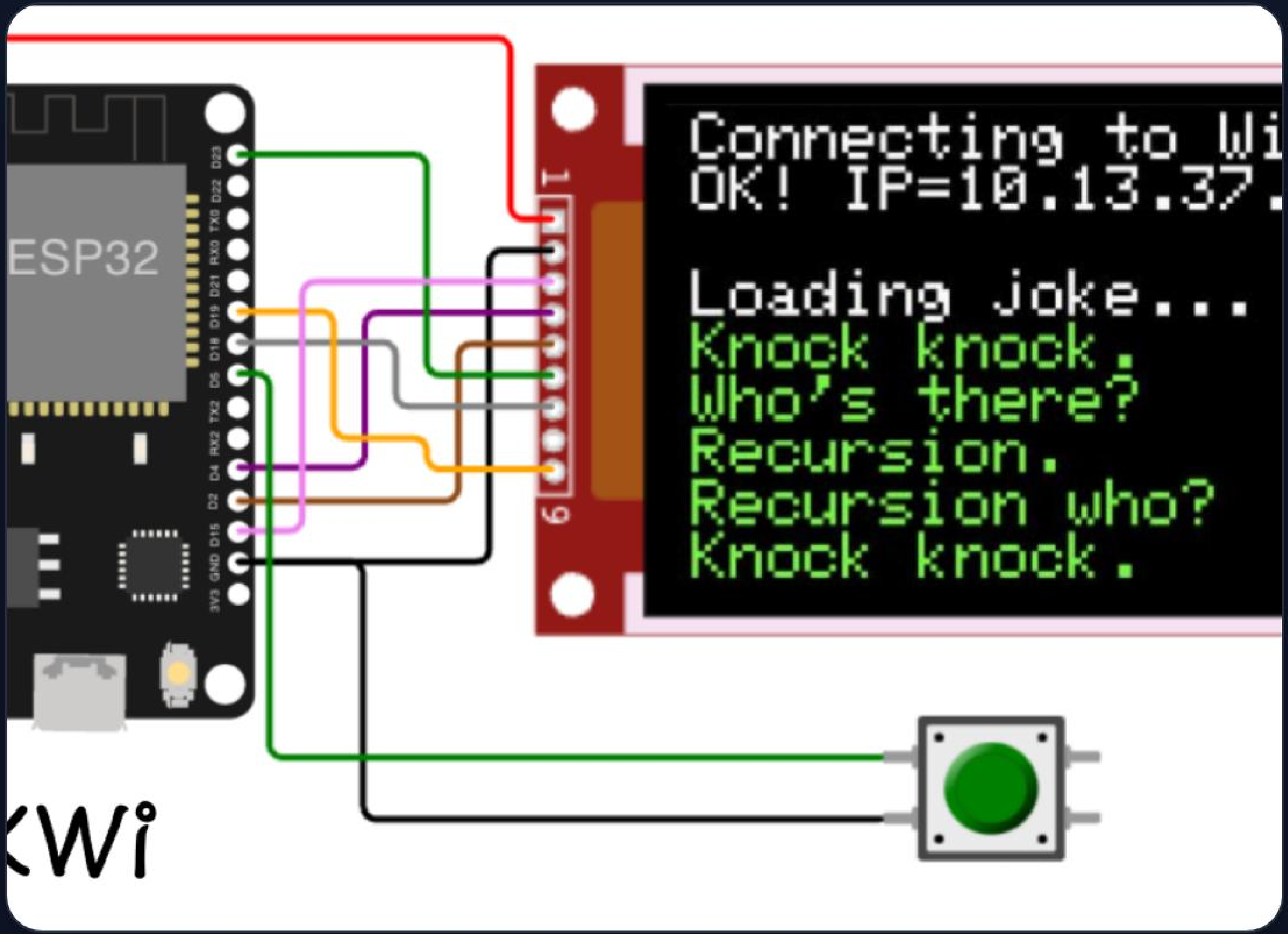
Head Pose

Detects distraction or head tilting associated with falling asleep.

FATIGUE SCORING MODEL

SCORE RANGE	DRIVER STATE	SYSTEM ACTION
0 – 74	Normal	Green Heartbeat (Safe)
75 – 84	Warning	Yellow LED + Slow Beep
85 – 89	Alarm	Orange LED + Fast Buzzer
90 – 100	Critical / Micro-sleep	Red LED + SOS Countdown

EMBEDDED SUBSYSTEM



Key Components

- **ESP32:** Main controller & logic unit.
- **MPU6050:** Accelerometer for crash detection.
- **GPS Module:** Real-time coordinate logging.
- **LCD 16×2:** Visual driver feedback.
- **Buzzer/LEDs:** Progressive alert system.

SYSTEM STATE MACHINE

NORMAL

Safe baseline; Green heartbeat active.

WARNING / ALARM

Increasing risk; Visual and audio cues.

EMERGENCY

Critical failure; Siren + SOS transmission.

CRASH DETECTION LOGIC

3.2G

Typical Crash Threshold

Impact Analysis

Uses MPU6050 Acceleration and Gyroscope data to filter noise from actual impacts.

- Force Spike Correlation
- Angular Rotation Check
- False-Positive Reduction Counter

EMERGENCY SOS WORKFLOW

Detection
Impact or micro-sleep confirmed

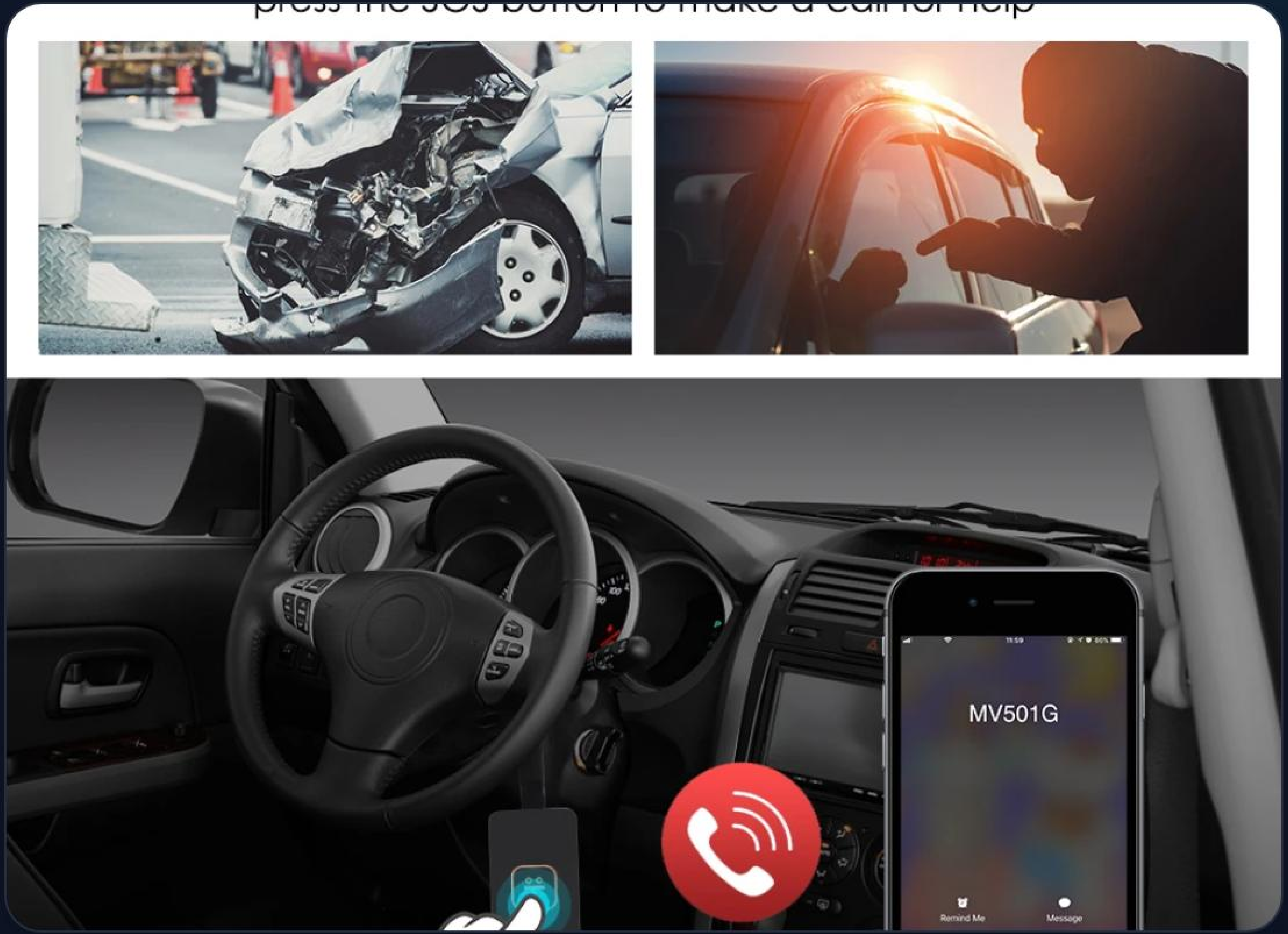
Countdown
T-minus 10s to manual cancel

GPS Pull
Fetch Lat/Long coordinates

Transmission
Simulated SMS via AT Commands

SOS MESSAGE PAYLOAD

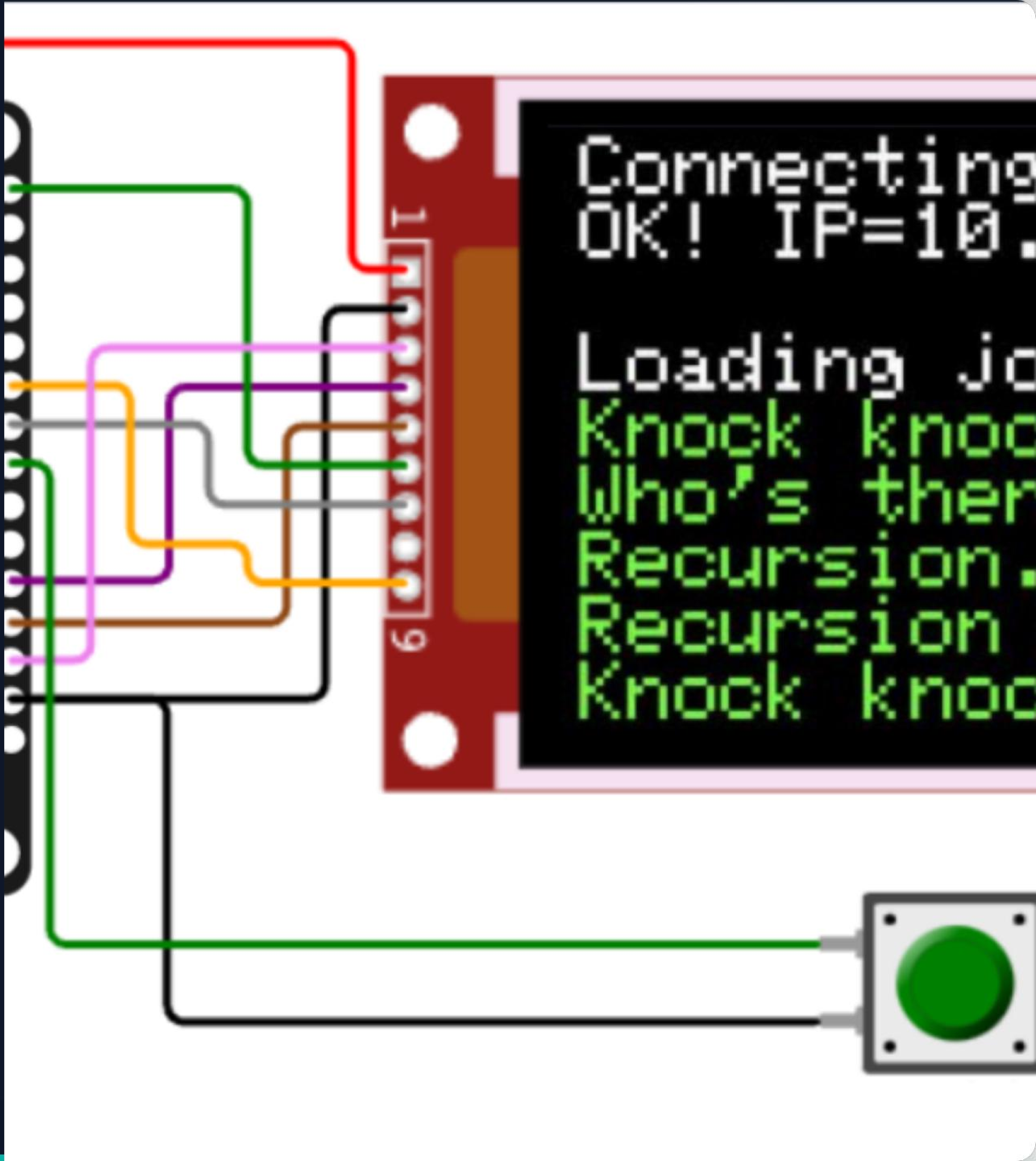
> *EMERGENCY ALERT!*
> *Cause: CRASH IMPACT*
> *Fatigue: 92% | Speed: 86 km/h*
> *Location: 9.0301, 38.7613*
> *Google Maps: http://maps.google.com/?q=9.0301,38.7613*



WOKWI SIMULATION

The system was successfully implemented and validated in the **Wokwi Virtual Environment**.

ESP32 DevKit simulation allows for rapid state machine testing without physical hardware constraints during the design phase.



ADMIN DASHBOARD CONCEPT

Fleet Safety Management

Production-grade concept for central monitoring:

- Live KPI Cards (Alerts/Fatigue)
- Driver Camera Simulation
- Real-time GPS Tracking

DEMONSTRATION SCENARIOS

1. Normal Driving

Stable metrics; LCD shows "System OK"; Green LED heartbeat pulse.

2. Escalating Fatigue

Potentiometer > 85%; Transitions from Warning to Alarm buzzer patterns.

3. Critical Crash

Manual crash button triggered; Red LED flash; SOS message printed to serial.

4. False Alarm Mute

User presses "Cancel" during countdown; System resets to Normal.

TESTING & CALIBRATION

METRIC	THRESHOLD VALUE	CONDITION TESTED
EAR (Eye Ratio)	< 0.25	Drowsiness / Eye Closure
PERCLOS	> 20%	Sustained fatigue over time
G-Force	> 3.0 G	Sudden impact / Collision
SOS Timer	10 Seconds	Countdown before message send

PROJECT ACHIEVEMENTS

Firmware

Developed robust ESP32 state machine with telemetry JSON output.

CV Design

Integrated MediaPipe and EAR for reliable driver state estimation.

Academic

Validated preventive and reactive safety logic in a virtual ecosystem.

CHALLENGES & LIMITATIONS

Technical Hurdles

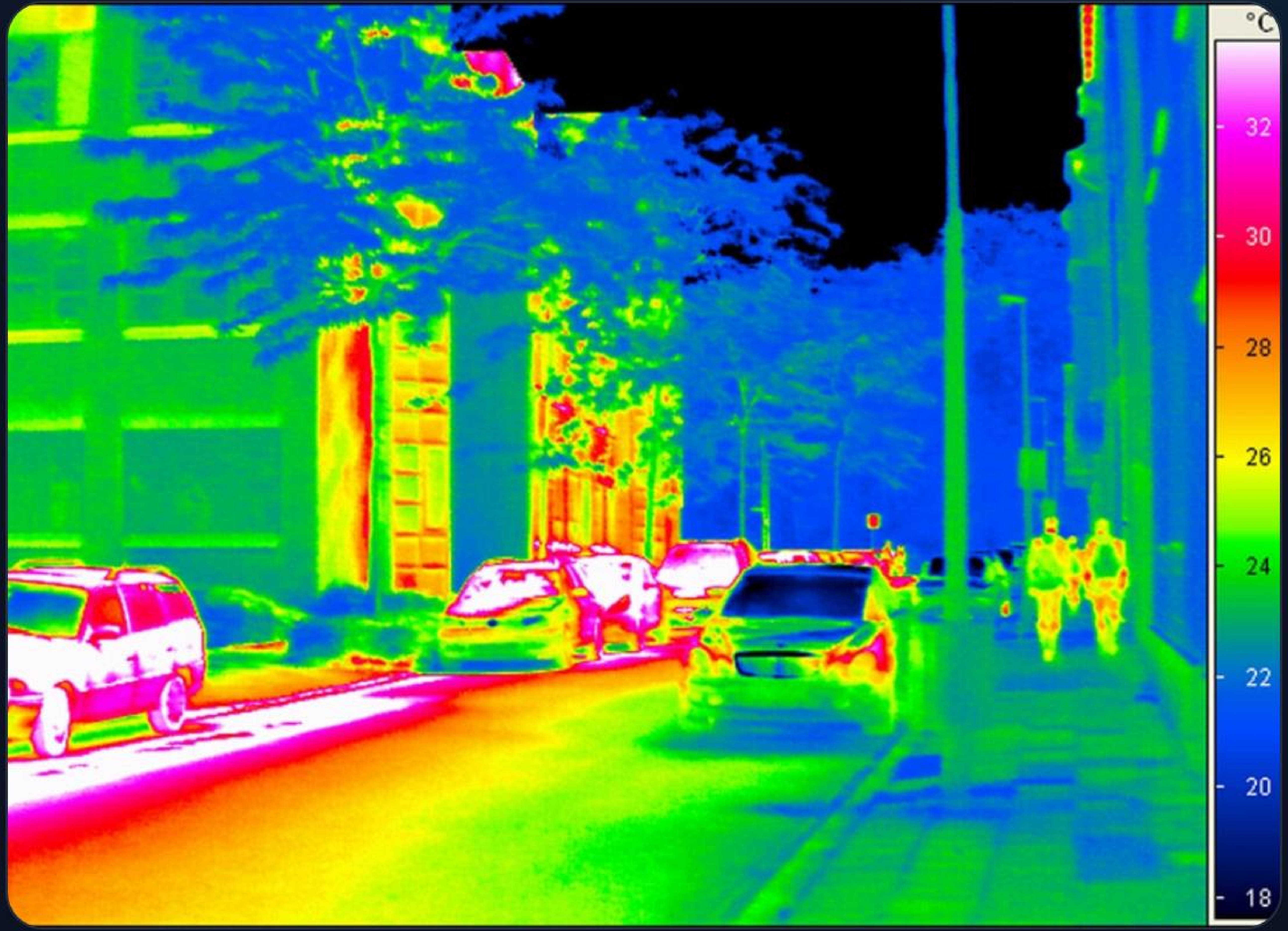
- CV sensitivity to poor lighting
- Obstructions from eyewear
- Simulated GSM vs Real Network

Simulation Scope

Current prototype is virtual. Physical automotive certification requires rigorous road-testing and vibration shielding for sensors.

FUTURE ROADMAP

- **Physical Prototyping:** Move to real ESP32 and SIM800L modules.
- **IR Camera:** Enable infrared monitoring for night-time safety.
- **Cloud Sync:** Fleet-wide safety reporting via MQTT/Cloud.

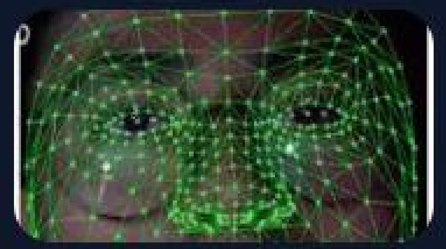


Conclusion & Q&A

Our two-tier safety system provides a scalable, low-cost solution to bridge the gap between driver monitoring and emergency response.

Thank you for your attention.

IMAGE SOURCES



https://www.mdpi.com/sensors/sensors-26-00889/article_deploy/html/images/sensors-26-00889-g002.png

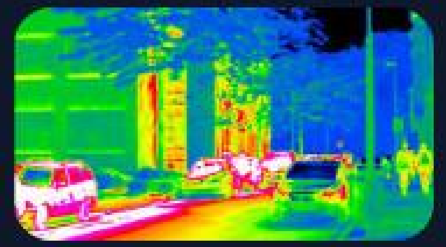
Source: www.mdpi.com

https://wokwi.com/images/social-cover.png?v=5

Source: wokwi.com

https://ae-pic-a1.aliexpress-media.com/kf/S54ffedbed5c34d5b815fffd7d9328d870.jpg

Source: www.aliexpress.com



https://s1.cdn.autoevolution.com/images/news/don-t-check-night-vision-for-drivers-infrared-cameras-22199_1.jpg

Source: www.autoevolution.com